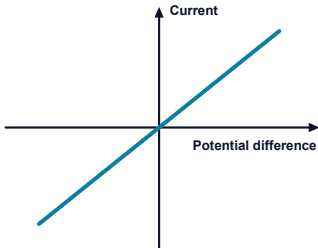
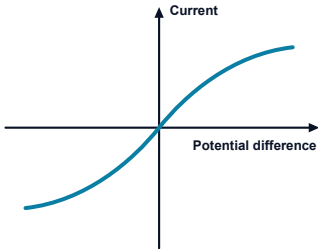
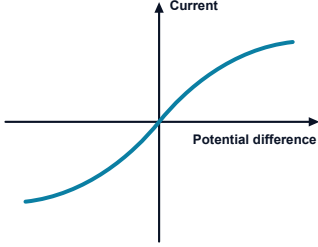
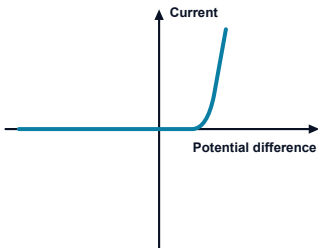


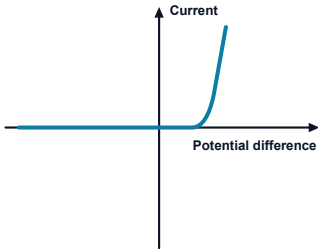
Current, resistance and potential difference

Electricity
Paper 1 · Combined and Separate
24 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

	QUESTION	ANSWER	✓
1	What is an electrical conductor?	A material that allows charge to flow through it.	☐
2	Name the charged subatomic particles that can flow in metals.	Electrons	☐
3	Define electric current.	The rate of flow of electrical charge.	☐
4	State the SI units of charge, current and time.	Charge: coulomb (C) Current: ampere (A) Time: second (s)	☐
5	Describe the current at different points in a single closed loop.	It is the same at every point.	☐
6	Name the two variables that the current through a component depends on.	The potential difference across it and its resistance	☐
7	The potential difference across a fixed resistor is increased. Describe the effect on the current.	The current increases.	☐
8	The resistance of a component is increased at a constant potential difference. Describe the effect on the current.	The current decreases.	☐
9	State the SI units of potential difference and resistance.	Potential difference: volt (V) Resistance: ohm (Ω)	☐
10	Name the type of component whose resistance stays constant as the current changes.	An ohmic conductor (at constant temperature)	☐
11	Describe how current depends on potential difference for an ohmic conductor.	Current is directly proportional to potential difference.	☐

QUESTION	ANSWER	✓
12 Name the type of component that gives this I–V graph. 	An ohmic conductor (e.g. a fixed resistor)	☐
13 Name the component that gives this I–V graph. 	A filament lamp	☐
14 This is the I–V graph for a filament lamp. Describe how its resistance changes as the potential difference increases. 	The resistance increases.	☐
15 Explain why the resistance of a filament lamp increases with current.	A greater current makes the filament hotter, and its resistance increases with temperature.	☐
16 Name the component that gives this I–V graph. 	A diode	☐

QUESTION		ANSWER	✓
17	This is the I–V graph for a diode. Describe its resistance when the potential difference is reversed.  The graph shows Current on the vertical axis and Potential difference on the horizontal axis. The curve is a horizontal line along the negative potential difference axis, indicating zero current flow in reverse bias. In forward bias (positive potential difference), the current remains at zero until a threshold voltage is reached, after which it increases exponentially.	Very high (almost no current flows).	☐
18	Describe how the resistance of a thermistor varies with temperature.	As the temperature increases, the resistance decreases.	☐
19	Describe how the resistance of an LDR varies with light intensity.	As the light intensity increases, the resistance decreases.	☐
20	Give one use of a thermistor and one use of an LDR.	Thermistor: thermostat (temperature sensor). LDR: switching lights on when it gets dark (light sensor).	☐
21	State the two things needed for electrical charge to flow in a circuit.	A complete circuit and a source of potential difference.	☐
22	State the equation linking charge flow, current and time.	charge flow = current $\times$ time ($Q = I t$)	☐
23	State the equation linking potential difference, current and resistance.	potential difference = current $\times$ resistance ($V = I R$)	☐
24	State the condition for the current through a resistor to be directly proportional to the potential difference.	Its temperature must stay constant.	☐